

Topic 11

MRRH (2018): The Model

Gabriel M Ahlfeldt

Quantitative Spatial Economics

Humboldt University & Berlin School of Economics
Summer term 2026

Acknowledgements

- ▶ This slide deck uses material from the following lectures
 - ▶ Lecture 5 (Commuting, Migration, and Local Employment Elasticities) of Spatial Economics Lectures—by Stephen J. Redding

Introduction

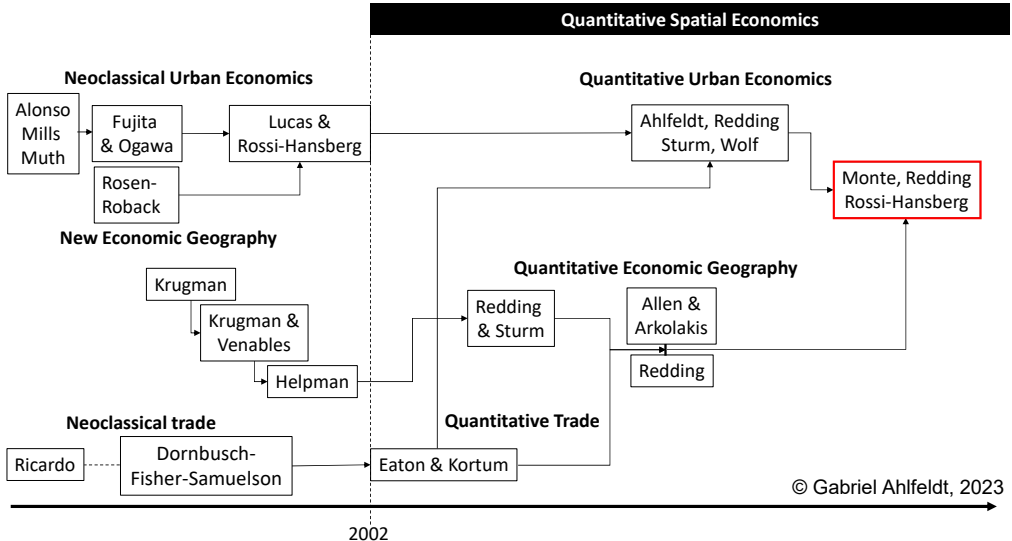
Recall

- ▶ **Quantitative urban models**
 - ▶ Focus on spatial linkages between locations in factor markets (commuting)
- ▶ **Quantitative economic geography models**
 - ▶ Focus on spatial linkages between locations in goods markets (trade)
- ▶ Many **similarities**
 - ▶ Realistic geography with discrete locations of arbitrary shape
 - ▶ Gravity equations & market access
 - ▶ Concentration and dispersion forces & spatial competition
 - ▶ Rationalize observed data via structural residuals (fundamentals)
 - ▶ Estimation, inversion, counterfactuals

Monte, Redding, Rossi-Hansberg (2018) (MRRH)

- ▶ **Commuting and trade linkages in one model**
 - ▶ A 'true' **quantitative spatial model**?
- ▶ **Resolves tension** in quantitative economic geography models
 - ▶ If units are **too large** $\Rightarrow$ Many local effects are not detectable
 - ▶ E.g. localized effects of highways or subways
 - ▶ If units are **too small** $\Rightarrow$ Underestimate local labour supply
 - ▶ Miss that productive locations can draw workers from other locations
- ▶ **Spatial equilibrium**
 - ▶ Concentration force: Home market effect (like in RRH)
 - ▶ No productivity and amenity spillovers (could be added)
 - ▶ Dispersion forces: Inelastically supplied land & idiosyncratic amenity (like in ARSW)
 - ▶ Zero profits and constant utility (like in ARSW & RRH)

History of thought



Roadmap

- ▶ **Monte, Redding, Rossi-Hansberg (2018)**
 - ▶ A canonical QSM
- ▶ **Topic 11 (today)**
 - ▶ Model
 - ▶ Equilibrium
 - ▶ Quantification
- ▶ **Topic 12**
 - ▶ Seidel & Wickerath (2020) version of the model
 - ▶ Application to Germany
 - ▶ Toolkit written in MATLAB (instead of Mathematica) **MRRH2015-toolkit**
 - ▶ Counterfactuals

Buidling blocks

Selection of building blocks (RRH2017 typology)

1. **Preferences:** Love of variety; Single traded sector; **workplace and residence amenities**; Residential land use; **idiosyncratic preferences for commuting routes**
2. **Production technology:** Increasing returns to scale; Exogenous productivity; No input-output linkages; No commercial land use
3. **Technology for the Movement of Goods:** Iceberg variable trade costs; **arbitrary** trade costs; Economic and Geographic Frictions; No non-traded goods besides residential land use
4. **Technology for the Movement of Ideas:** No knowledge externalities or diffusion; No innovation; No transferability of ideas
5. **Technology for the Movement of People:** Perfectly costless migration; **commuting**; Single worker type with no heterogeneity; No congestion in transportation
6. **Endowments:** Homogenous labor; Exogenous land endowments in regions within a single country; No capital
7. **Equilibrium:** Monopolistic competition; General equilibrium with a single country; **immobile landlords consuming income locally**; Trade is balanced in each location

A Canonical Quantitative Spatial Model

- ▶ Consider an economy consisting of a set N of **regions indexed by n**
 - ▶ Each region is endowed with an **exogenous supply of land** (H_n)
- ▶ Economy is endowed with a **measure $\bar{L}$ of workers** who supply one unit of labor
- ▶ Workers are **perfectly geographically mobile**
- ▶ Regions are connected by a **bilateral transport network**
 - ▶ Can be used to **ship goods subject to iceberg trade costs**
 - ▶ $d_{ni} = d_{in} > 1$ units must be shipped from region i in order for one unit to arrive in region $n \neq i$; $d_{nn} = 1$
 - ▶ Can be used to **commute** subject to an iceberg commuting costs
 - ▶ $\kappa_{ni} \in [1, \infty)$ commuting cost in terms of utility (proportionate to opportunity cost of time)

Preferences and Amenities (similar to ARSW)

- ▶ Utility of worker ω that lives in n and works in i is

$$U_{ni\omega} = \frac{b_{ni\omega}}{\kappa_{ni}} \left(\frac{C_{n\omega}}{\alpha} \right)^\alpha \left(\frac{H_{n\omega}}{1-\alpha} \right)^{1-\alpha}$$

- ▶ where $C_{n\omega}$ is the CES consumption basket with elasticity of substitution σ (like in RRH), and $H_{n\omega}$ housing consumption
- ▶ Utility **cost of commuting** are given $\kappa_{ni} \in [1, \infty)$
 - ▶ Since it enters multiplicative in utility, it is proportionate to wage
- ▶ Amenities, $b_{ni\omega}$, drawn i.i.d. from Fréchet distribution $G_{ni}(b) = e^{-B_{ni}b^{-\epsilon}}$
 - ▶ where $B_{ni} > 0$ is the **average bilateral amenity (instead of T and E in ARSW)**
 - ▶ $\epsilon > 1$ governs the dispersion of idiosyncratic tastes

Production (similar to RRH)

- ▶ Horizontally **differentiated varieties sold under monopolistic competition**
- ▶ **Only labor** required to produce $x_i(j)$ units of output in i is

$$l_i(j) = F + \frac{x_i(j)}{A_i}$$

- ▶ Prices (constant markup) at n are given by $p_{ni}(j) = \left(\frac{\sigma}{\sigma-1}\right) \frac{d_{ni}w_i}{A_i}$,
 - ▶ where $d_{ni} \geq 1$ denotes iceberg transport costs between i and n
- ▶ Constant equilibrium output $x_i(j) = A_i F(\sigma - 1)$ implies measure of varieties

$$M_i = \frac{L_i}{\sigma F}, \text{ since } l_i(j) = \bar{l} = \sigma F$$

Land Market (sort of similar to RRH)

- ▶ Price of land Q_n determined from land market clearing
 - ▶ There is an **inelastic supply** of housing at H_n (**imperfectly elastic in extension**)

$$\underbrace{H_n Q_n}_{\text{housing supply}} = \underbrace{(1 - \alpha) v_n R_n}_{\text{housing demand}} \Rightarrow Q_n = (1 - \alpha) \frac{v_n R_n}{H_n},$$

- ▶ where v_n is expected income of residents at n and R_n is the total number of residents
- ▶ Land owned by **landlords**, who receive income from residents' expenditure on land
 - ▶ **Consume goods where they live**
- ▶ Total expenditure on goods is the sum of expenditures by residents and landlords

$$P_n C_n = \alpha v_n R_n + (1 - \alpha) v_n R_n = v_n R_n$$

Trade in Goods (as in RRRH)

- ▶ Expenditure shares are given by

$$\pi_{ni} = \frac{L_i (d_{ni} \frac{w_i}{A_i})^{1-\sigma}}{\sum_{k \in N} L_k (d_{nk} \frac{w_k}{A_k})^{1-\sigma}}$$

- ▶ Recall: This is the expenditure agents in n spend on goods produced in i
- ▶ The price of the consumption basket at n is given by

$$P_n = \left(\frac{\sigma}{\sigma - 1} \right) \left(\frac{L_n}{\sigma F \pi_{nn}} \right)^{\frac{1}{1-\sigma}} \frac{w_n}{A_n}$$

Work-Residence Decision (similar to ARSW)

- ▶ The indirect utility of an agent ω that lives in n and works in i is

$$U_{ni\omega} = \frac{b_{ni\omega} w_i}{\kappa_{ni} P_n^\alpha Q_n^{1-\alpha}}$$

- ▶ which is drawn from $G_{ni}(u) = e^{-\Psi_{ni} u^{-\epsilon}}$, with $\Psi_{ni} = B_{ni} (\kappa_{ni} P_n^\alpha Q_n^{1-\alpha})^{-\epsilon} w_i^\epsilon$
- ▶ The unconditional probability that a worker lives in n and works in i is

$$\lambda_{ni} = \frac{B_{ni} (\kappa_{ni} P_n^\alpha Q_n^{1-\alpha})^{-\epsilon} w_i^\epsilon}{\sum_{r \in N} \sum_{s \in N} B_{rs} (\kappa_{rs} P_r^\alpha Q_r^{1-\alpha})^{-\epsilon} w_s^\epsilon}$$

- ▶ Free mobility implies that $\bar{U} = E[U_{ni\omega}]$ for all ni

Commuting (similar to ARSW)

- ▶ Conditional probability that worker in n commutes to location i is

$$\lambda_{n|i} = \frac{B_{ni} \left(\frac{w_i}{\kappa_{ni}} \right)^\epsilon}{\sum_{s \in N} B_{ns} \left(\frac{w_s}{\kappa_{ns}} \right)^\epsilon}$$

- ▶ So labor market clearing implies that

$$L_i = \sum_{n \in N} \lambda_{n|i} R_n$$

- ▶ Expected residential income is then

$$v_n = \sum_{i \in N} \lambda_{n|i} w_i$$

Equilibrium

General Equilibrium

- ▶ The general equilibrium can be referenced by a vector of
 - ▶ six variables $\{w_n, v_n, Q_n, L_n, R_n, P_n\}$
 - ▶ and a scalar U
- ▶ This vector solves
 - ▶ Seven equations
 - ▶ Income equals expenditure (7): $w_i L_i = \sum_{n \in N} \pi_{ni} \bar{v}_n R_n$
 - ▶ average residential income (14): $\bar{v}_n = \sum_{i \in N} \lambda_{ni}^R w_i$
 - ▶ choice probabilities (2x 11): $\lambda_n = \sum_{i \in N} \frac{\phi_{ni}}{\Phi}, \quad \lambda_i = \sum_{n \in N} \frac{\phi_{ni}}{\Phi}$
 - ▶ price indices (8): $P_n = \frac{\sigma}{\sigma-1} \left(\frac{L_n}{\sigma F \pi_{nn}} \right)^{\frac{1}{1-\sigma}} \frac{d_{nn} w_n}{A_n}$
 - ▶ Two labour market clearing condition $\bar{L} = \sum_{n \in N} R_n = \sum_{n \in N} L_n$
- ▶ **7 endogenous objects, 7 equations ✓**
 - ▶ MRRH provide sufficient conditions for equilibrium uniqueness and existence

Quantification

MRRH Data for Calibration

- ▶ Commodity Flow Survey (CFS)
 - ▶ Bilateral trade between 123 CFS regions
 - ▶ Bilateral distance shipped
- ▶ American Community Survey (ACS)
 - ▶ Commuting probabilities between counties
- ▶ Bureau of Economic Analysis
 - ▶ Employment by workplace county
 - ▶ Wages by workplace county
- ▶ GIS data
 - ▶ County maps
- ▶ Parameters from literature
 - ▶ Elasticity of substitution, $\sigma = 4$ (Broda and Weinstein, 2006),

Setting the Expenditure share on housing

- ▶ The **non-housing expenditure share**:
 - ▶ In RRH2017: $\alpha = 0.75$ (Davis and Ortalo-Magne, 2011)
 - ▶ In MRRH2018: $\alpha = 0.6$ (Bureau of Economic Analysis)
- ▶ Recall: Sufficient condition for uniqueness from RRH (exogenous housing supply)
 - ▶ $\sigma(1 - \alpha) > 1$
- ▶ MRRH have **endogenous housing supply in model extensions**
 - ▶ Congestion force is smaller $\Rightarrow$ Risk of **multiple equilibria**
 - ▶ Helpful if that sufficient statistic is higher...

	σ	α	$\sigma(1 - \alpha) > 1$
RRH	5	0.75	1.25
MRRH	4	0.6	1.6

Estimating distance elasticity of trade

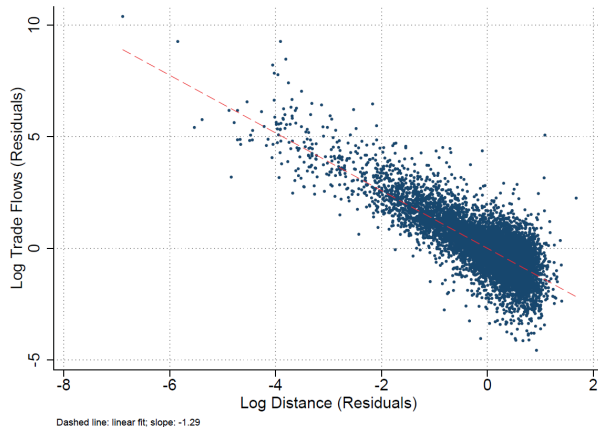
- ▶ MRRH **quantify the model for US counties**
 - ▶ Only observe **trade for CFS regions, which are larger than counties**
- ▶ Trade data still helps **estimating distance elasticity of trade** $\psi(1 - \sigma)$

$$\ln \pi_{ni} = [(1 - \sigma)\psi] \ln \text{dist}_{ni} = O_i + D_n + [(1 - \sigma) \ln e_{ni}],$$

- ▶ where ψ is the distance elasticity of trade cost
 - ▶ and $\ln e_{ni}$ captures unobserved trade costs
 - ▶ Standard empirical trade gravity equation (see Topic 9)
- ▶ RRH set $[(1 - \sigma)\psi] = -1.5$
 - ▶ Ahlfeldt, Roth, Seidel (2023) estimate $[(1 - \sigma)\psi] = -3$ for Germany (see Topic 9)

Estimating distance elasticity of trade

- ▶ Estimated slope:
 $[(1 - \sigma)\psi] = -1.29$
 - ▶ After controlling for origin and destination fixed effects
 - ▶ Implies $\psi = 0.43$ given $1 - \sigma = 3$
 - ▶ Log-log approximation not too bad
- ▶ Can predict $d_{ni} = \text{dist}_{ni}^{\psi}$ at county level



Inversion of Productivities

- ▶ County trade balance (exports and imports including to itself) implies

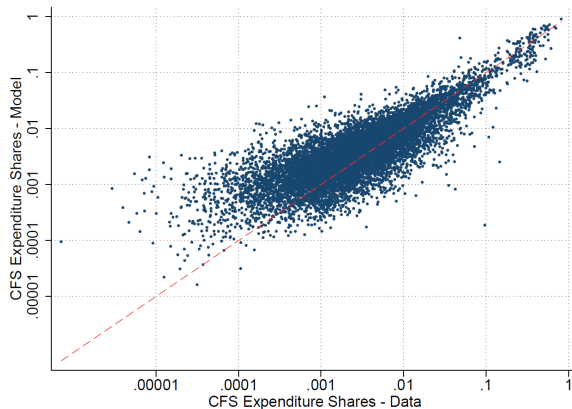
$$w_i L_i = \sum_{n \in N} \pi_{ni} v_n R_n = \sum_{n \in N} \frac{L_i (d_{ni} w_i)^{1-\sigma} A_i^{\sigma-1}}{\sum_{k \in N} L_k (d_{nk} w_k)^{1-\sigma} A_k^{\sigma-1}} v_n R_n.$$

- ▶ We observe (or can compute) $\{w_i, L_i, R_i, v_i\}$
- ▶ We have predicted $d_{ni}^{1-\sigma} = (\text{dist}_{ni})^{-1.29}$
- ▶ We can (numerically) solve uniquely for productivities, A_i (up to a constant)
 - ▶ Intuitively, firms in i pay for wage bill with expenditures they receive from n
 - ▶ If $w_i L_i > \sum_{n \in N} \pi_{ni} v_n R_n$, **not enough expenditure on goods produced in i**
 - ▶ **Need to increase A_i** $\Rightarrow$ productivity increases, price falls and production expands
 - ▶ Expenditure shares on goods from i increase $\Rightarrow$ Revenues of firms in i increase
 - ▶ Keep adjusting A_i until revenues $\sum_{n \in N} \pi_{ni} v_n R_n$ exactly match $w_i L_i$
- ▶ More next next week... MRRH2018-toolkit

Overidentifying trade shares

- ▶ How well do we do on bilateral shares?
 - ▶ Model generates county-shares
$$\pi_{ni} = \frac{L_i (d_{ni} \frac{w_i}{A_i})^{1-\sigma}}{\sum_{k \in N} L_k (d_{nk} \frac{w_k}{A_k})^{1-\sigma}}$$
 - ▶ Aggregate model shares to CFS level
 - ▶ Model shares **positively correlated** with observed CFS shares

Does it look alright?



Estimating commuting decay

- The unconditional probability that a worker lives in n and works in i is

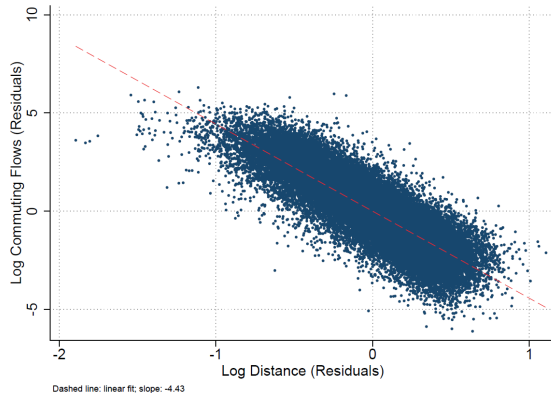
$$\lambda_{ni} = \frac{B_{ni} (\kappa_{ni} P_n^\alpha Q_n^{1-\alpha})^{-\epsilon} w_i^\epsilon}{\sum_{r \in N} \sum_{s \in N} B_{rs} (\kappa_{rs} P_r^\alpha Q_r^{1-\alpha})^{-\epsilon} w_s^\epsilon}$$

- Let $\log \mathcal{B}_{ni} = \log(B_{ni} \kappa_{ni}^{-\epsilon}) = \log \mathbb{B}_n + \log \mathbb{B}_i - \phi(\log(\text{dist}_{ni})) + \log \mathbb{B}_{ni}$,
 - where implicitly $\kappa_{ni} = \text{dist}_{ni}^\tau$ so that $-\phi \equiv -\epsilon\tau$
- We can estimate ϕ in a reduced-form regression

$$\log \lambda_{ni} = g_0 + \eta_n + \mu_i - \phi \log \text{dist}_{ni} + \log \mathbb{B}_{ni},$$

Commuting decay

- ▶ Estimated commuting decay
 $-\phi = -4.43$
- ▶ Decay is much steeper than
in trade (-1.29)
- ▶ Expected, since **moving
people is more expensive
than moving goods**



Estimating preference heterogeneity

- ▶ To estimate the preference heterogeneity ϵ , remove w_i from destination FE

$$[\log \lambda_{ni} + \phi \log \text{dist}_{ni}] = g_0 + \eta_n + \epsilon \log w_i + \log u_{ni},$$

- ▶ where $\log u_{ni} = \log \mathbb{B}_i + \log \mathbb{B}_{ni}$ and ϕ taken from previous regression
- ▶ w_i **correlated with error** component $\mathbb{B}_i$, which is a labour supply shifter
 - ▶ Workers accept lower wage if there is a positive workplace amenity
- ▶ Need an **instrument** for wage that is uncorrelated with labour supply
 - ▶ Use fundamental productivity A_i as a labour demand shifter $\Rightarrow \epsilon = 3.3$
 - ▶ Have inverted A_i using *income = expenditure* and observed $\{w_i, L_i, d_{ni}\}$

Q: Sounds good?

Inversion of amenities

- ▶ We observe or can solve for all elements of bilateral commuting probabilities

$$\lambda_{ni} = \frac{B_{ni} (\kappa_{ni} P_n^\alpha Q_n^{1-\alpha})^{-\epsilon} w_i^\epsilon}{\sum_{r \in N} \sum_{s \in N} B_{rs} (\kappa_{rs} P_r^\alpha Q_r^{1-\alpha})^{-\epsilon} w_s^\epsilon}$$

- ▶ except 'ease of commuting' $\mathcal{B}_{ni} = (B_{ni} \kappa_{ni}^{-\epsilon})$
 - ▶ Straightforward to recover $\mathcal{B}_{ni}$ as the structural residual
- ▶ **$\mathcal{B}_{ni}$ is all we need to run counterfactuals**
 - ▶ Since B_{ni} and κ_{ni} are isomorphic in the model,
- ▶ Notice that amenities are essentially zero on routes without flows
 - ▶ No flows on these routes in counterfactuals
 - ▶ Not recommended in granular settings (Dingle and Tintelnot, 2026)

Counterfactuals

MRRH counterfactuals

- ▶ MRRH use the model to perform different counterfactuals
 - ▶ Local employment elasticities
 - ▶ Changes in commuting costs
- ▶ Derived in terms of exact hat algebra

We discuss the MRRH counterfactuals briefly and then move on to the MRRH2018-toolkit and its didactic counterfactuals due to accessibility of the MATLAB code [MRRH2018-toolkit](#)

Exact hat algebra

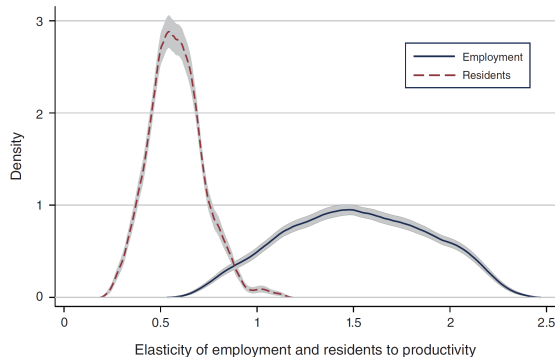
- ▶ MRRH show how to use exact hat algebra in numerical solver to obtain changes in endogenous variables for given
 - ▶ changes in exogenous variables $\{\hat{A}_n, \hat{B}_n, \hat{\kappa}_{ni}, \hat{d}_{ni}\}$
 - ▶ Values structural parameters $\{\alpha, \sigma, \epsilon, \delta, \kappa\}$
 - ▶ Observed data $\{L_{ni}, w_i, \pi_{ni}\}$
- ▶ Treat $\{\hat{\lambda}_{ni}, \hat{w}_i\}$ as target variables
 - ▶ Start with guesses
 - ▶ Use recursive structure to solve for values of other referencing objects
 - ▶ Use values of other referencing objects to predict $\{\hat{\lambda}_{ni}, \hat{w}_i\}$
 - ▶ Update $\{\hat{\lambda}_{ni}, \hat{w}_i\}$ until convergence
- ▶ More next week... **MRRH2018-toolkit**

Local employment elasticities

- ▶ Find out how strongly **employment reponds to labour demand shocks**
 - ▶ Local employment elasticity $\equiv$ Elasticity of employment with respect to productivity
- ▶ Key point is that extant literature underestimates this elasticity
 - ▶ By focusing on migration and **ignoring commuting**
 - ▶ Commuting makes local labor supply much more elastic
- ▶ **Conduct 3,111 counterfactuals** (one for each county)
 - ▶ Shock a county by 5%
 - ▶ Use the model to compute by how much local employment changes
 - ▶ Compute the local employment elasticity
- ▶ Compare to **residents elasticity** (how does resident pop. respond)

Local employment elasticities

- ▶ Employment elasticity ranges from 0.5 to 2.5 with mean of 1.52
- ▶ Resident elasticity much lower (0.2-1.2)
- ▶ Employment elasticity is larger in "open" commuting markets
 - ▶ Substantiated by reduced-form evidence



Commuting cost

- ▶ What are the effects of empirically observed **reductions in commuting cost**?
 - ▶ Due to improvements in rolling stock and infrastructure
- ▶ Measure ease of commuting in relative terms: $\tilde{\mathcal{B}}_{ni} \equiv \left(\frac{\mathcal{B}_{ni}\mathcal{B}_{in}}{\mathcal{B}_{nn}\mathcal{B}_{ii}} \right)^{\frac{1}{2}} = \left(\frac{L_{ni}L_{in}}{L_{nn}L_{ii}} \right)^{\frac{1}{2}}$
 - ▶ **Normalize bilateral 'ease' by within 'ease'** at origin and destination
 - ▶ Ease of commuting only identified up to a constant
 - ▶ Geometric mean of ease in both directions (assuming symmetry of infrastructure?)
- ▶ Can express 'ease of commuting' solely in terms of observed commuting flows L_{ni}
 - ▶ All other terms in the commuting probability equation cancel out after normalization
 - ▶ Recall that $L_{ni} = \lambda_{ni}\bar{L}$

Commuting cost counterfactual

- ▶ Relative ease of commuting increased from 1990 to 2010
 - ▶ by 4 (25th pct) to 12 (50th pct) and 21 (75th pct) percent
- ▶ Counterfactuals where $B_{ni}\kappa_{ni}^{-\epsilon}$ change on all routes
 - ▶ Change primitives, simulate effects on endogenous outcomes, and plug into:

$$\hat{U} = \left(\frac{1}{\hat{\lambda}_{ij}} \right)^{\frac{1}{\epsilon}} \left(\frac{1}{\hat{\pi}_{ij}} \right)^{\frac{\alpha}{\sigma-1}} \left(\frac{\hat{w}_i}{\hat{v}_i} \right)^{1-\alpha} \frac{\hat{L}_i^{\frac{\alpha}{\sigma-1}}}{\hat{R}_i^{1-\alpha}}$$

- ▶ Commuting **cost reductions have increased utility** by 0.9 to 6.9 percent
 - ▶ Assuming observed changes in ease of commuting are driven by κ (not B)
 - ▶ No change in internal ease of commuting...

Conclusion

Summary

- ▶ **Quantitative urban models** emphasize the trade between regions
 - ▶ Follow in the tradition of neoclassical urban models
- ▶ **Quantitative economic geography** emphasize the trade between regions
 - ▶ Follow in the tradition of New Economic Geography models
- ▶ **MRRH** integrate spatial linkages via **commuting and trade** into one model
 - ▶ Use the model to show that commuting makes local labour supply more elastic
 - ▶ Also show that improvements in commuting technology have generated welfare

Next week: **Counterfactuals for Germany**

Literature I

Core readings

- ▶ **Monte, F., Redding, S., Rossi-Hansberg, E. (2018): Commuting, Migration, and Local Employment Elasticities. *American Economic Review*, 108(12), 3855-90.**

Other readings

- ▶ Ahlfeldt, G., Redding, S., Sturm, D., Wolf, N. (2015): The economics of Density: Evidence From the Berlin Wall. *Econometrica*, 83(6), 2127–2189.
- ▶ Ahlfeldt, G. M., Roth, D., Seidel, T. (2023): Optimal minimum wages in spatial economies. Working paper.
- ▶ Broda, C., Weinstein, D. (2006): Globalization and the Gains From Variety. *The Quarterly Journal of Economics*, 121(2), 541–585.
- ▶ Davis, M., F. Ortalo-Magné (2011): Household expenditures, wages, rents, *Review of economic dynamics*, 14(2), 248-261.
- ▶ Dingle, J., Tintelnot, F. (2026): Spatial Economics for Granular Settings. *Econometrica*. 94(2), 407-464.
- ▶ Redding, S., Rossi-Hansberg, E. (2017): Quantitative Spatial Economics. *Annual Review of Economics*, 9, 21–58.